INCH-POUND

MIL-PRF-1851G
01 May 2023
SUPERSEDING
MIL-B-1851F
28 May 1981

PERFORMANCE SPECIFICATION

BAND, HELMET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of helmet band.

1.2 Classification. The helmet band will be of the following classes as specified (see 6.2):

Class 1 - Olive Drab 7
Class 2 - Tan 380

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129 - Military Marking For Shipment and Storage

(Copies of this document are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

2-2-420 Band, Helmet

(Copies of this drawing are available from the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted will be those listed in the issue of the DoDISS and the supplement thereto, if applicable.

ASTM International:

ASTM D6193 - Stitches and Seams, Standard Practice for.

(Copies of this document are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

American Society for Quality (ASQ):

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes..

(Copies of this document are available online from http://asq.org or from the ASQ Distribution Center, 600 North Plankinton Avenue, Milwaukee, Wisconsin 53207-6028.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification will take precedence.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case, this specification shall govern.

3.1.2. First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Materials. Material shall be as specified on Drawing 2-2-420.

3.3 Construction. The construction shall conform in all respects to Drawing 2-2-420 and as specified herein.

3.4 Stitches, seams and stitching. All stitching except bartacking shall be with 8 to 10 stitches per inch, using stitch type 301 conforming to ASTM D6193.

3.4.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched ½ inch minimum. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn.

3.4.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

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a. When thread breaks or bobbin runouts occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of ½ inch back of the end of the stitching.

b. Thread breaks, two or more consecutive skilled or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of ½ inch in back of the defective area, continue over the defective area and continue a minimum of ½ inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required area. When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.2 Automatic stitching. Automatic sewing machines may be used to perform the required stitch pattern provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.4.3 Bartacking. Bartacks shall be ½ ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose stitching.

3.4.4 Thread ends. All thread ends shall be trimmed to ¼ inch maximum length.

3.5 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damage that are classified as defects in 4.4.2.1 shall be removed from production and replaced with non-defective components.

3.6 Marking. The nomenclature shall be applied on the on the inside of each band in characters not less than ¼ in height, and shall conform to the requirements for type III or IV, class 9 of MIL-DTL-32075A.

3.7 Workmanship. The finished band shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. VERIFICATION

4.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. If required per 6.2, the first article sample shall be examined as specified in Table 1 below. Presence of one or more defects shall be cause for rejection of the item inspected.

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4.3.2 Visual Examination. The first article item shall be inspected as specified in Table 1. Failure of the examination shall be cause for rejection of helmet band assembly.

4.3.3 Dimensional examination. The helmet band shall be examined for all dimensions shown on Drawing 2-2-420. Any dimension exceeding the specified tolerance shall constitute a defect. The inspection level shall be S-3 and the AQL shall be 4.0 defects per hundred units.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with ASQ Z1.4, inspection level S-3.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in Table 1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Visual examination. The sample(s) shall be examined as specified herein, for the following defects:

TABLE 1

Examine	Defect	Classification Major	Classification Minor
Webbing	Any hole, cut or tear	X	
	Not firmly or tightly woven; edges frayed	X	
	Abrasion mark; smash	X	
	Broken or missing end or Pick		X
Thread	Color of needle thread for attaching luminescent tapes not natural		X
Luminescent tapes	Any hole, cut or tear	X	
	One or both omitted	X	
	Attached with luminescent side not facing out	X	
Seams and stitching (type 301 stitching)			
Open seams	Any	X	
	Note: a seam shall be classified as open when one or more stitches are broken or when two or more consecutive skipped or runoff stitches occur.		
Runoff (see open seams)			
Stitch type	Not as specified	X	
Stitch pattern	Any row of stitching omitted	X	
Stitch tension	Loose, resulting in loose or bobbin thread.		X

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	Excessively tight resulting in puckering or webbing		X
Stitches per inch	One stitch less than minimum specified		X
	Two or more stitches less than minimum specified	X	
	One or more stitches in excess of minimum specified		X
Stitching ends	Not secured as specified		X
Thread breaks, skipped stitches or runoffs	Overstitched less than ½ inch in each direction beyond the defective stitching area		X
	Note: Thread breaks, two or more consecutive or skipped or runoff stitches not overstitched shall be classified as open seams		
Bartacks (if used)	Any bartack omitted	X	
	Loose or broken stitching		X
Thread ends	Not trimmed to ¼ inch maximum length		X
Cleanness	Grease or oil stains clearly noticeable		X
Marking	Omitted, illegible, incorrect, misplaced, or size of characters not as specified		X

4.6 Inspection of packaging. The preservation, packing and marking for level A or B shall be examined to determine compliance with MIL–P-775. The preservation, packing and marking for commercial shall be examined in accordance with ASTM D3951.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The band is intended for use in conjunction with the combat helmet to support camouflage and to provide a means for detecting friendly personnel at night.

6.2 Ordering data. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval (see 3.2).

c. Degree of preservation and packing required (see 5.1 and 5.2).

d. Class (see 1.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model coupling assembly. The first article should consist of one or more coupling assemblies. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Subject term (key word) listing.

stitch type 301
bartack
olive drab 7
tan 380

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY ACTIVITIES:

Custodians:

Army – GL

Preparing activity:

DLA - CT

(Project 8415-2023-038)

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